Abhijit Shankar Mudigonda

Contact Information

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https://abhijit-mudigonda.github.io/math

Research Interests

Number Theory, Algebraic Geometry, Theoretical Computer Science

Education

University of Chicago

Ph.D. in Mathematics and Computer Science, September 2021 - June 2027
Advised by Frank Calegari and Janos Simon
M.S. in Computer Science, June 2024

Massachusetts Institute of Technology

B.S. in Mathematics, September 2015 - June 2019

Publications

A. S. Mudigonda. *Computing Hilbert modular forms with nonparitious weight*. Research in Number Theory, 2026. Proceedings of the Algorithmic Number Theory Symposium XVII, to appear.

B. Matschke and A. S. Mudigonda. *Quadratic Fields Admitting Elliptic Curves with Rational j-Invariant and Good Reduction Everywhere*. Journal of Number Theory. Volume 247, June 2023, Pages 162-210.

A. S. Mudigonda and R. R. Williams. *Time-Space Lower Bounds for Simulating Proof Systems with Quantum and Randomized Verifiers*. Proceedings of the 2021 Conference on Innovations in Theoretical Computer Science.

Preprints

A. S. Mudigonda. *Modularity of some atypical abelian fourfolds*.

E. Assaf, A. S. Mudigonda. *Computing Hilbert modular forms with nontrivial nebentypus*.

E. Assaf, A. Babei, B. Breen, S. Chari, E. Costa, J. Duque-Rosero, A. Horawa, J. Kieffer, A. Kulkarni, A. S. Mudigonda, M. Musty, S. Schiavone, S. Sethi, S. Tripp, J. Voight. *Computing with Fourier expansions of Hilbert modular forms*. In preparation.

Patents

D. Gutfreund, Q. Fan, A. Mudigonda. *Restructuring Deep Neural Networks to Reduce the Number of Parameters*. USPA #20200160144. Published May 21, 2020.

Invited Talks

January 2027 – JMM Special Session, Explicit Methods in Number Theory
January 2027 – JMM Special Session, Mathematics Informed by Computing
December 2026 – UW-Madison Number Theory Seminar
November 2026 – McGill Graduate Student Number Theory Seminar
October 2026 – University of Sydney Computational Algebra Seminar
July 2026 – Algorithmic Number Theory Symposium XVII
May 2026 – UChicago Number Theory Seminar
April 2026 – UC Santa Barbara Number Theory Seminar
January 2021 – Innovations in Theoretical Computer Science
January 2021 – UT Austin Quantum Information Seminar

Work Experience

08/2020 - 09/2021	*Software Engineer, Facebook Reality Labs* OS engineer on the XROS platform team, developing features for the Oculus Quest Pro.
01/2017 - 04/2017	*Research Intern, IBM Research.* Developed neural network architectures inspired by expander graphs.
06/2016 - 08/2016	*Research Intern, Knight Cancer Institute.* Used deep learning models to predict DNA sequencing accuracy.

Teaching Experience

09/2022- Present	*Teaching Assistant* Served as a TA for Honors Introduction to Discrete Mathematics, Introduction to Algorithms, Honors Introduction to Complexity Theory, Algebra I, Introduction to Riemannian Geometry, and Introduction to Algebraic Number Theory
09/2022 - 06/2023	*UChicago Teaching Fellow* Underwent a year-long training program preparing graduate students to be independent lecturers.
06/2020 - 08/2020	*Summer STEM Institute Research Mentor* Mentored two high school students in theoretical and applied quantum computing research projects.
01/2019	*MIT Global Teaching Labs Teacher* Taught physics and biology to high school students at the Liceo Scientifico Michelangelo Grigoletti in Pordenone, Italy.
06/2015 - 06/2018	*USA Biology Olympiad Teaching Assistant* Wrote practical and theoretical examinations to select the 2015, 2016, 2017, and 2018 U.S. representative teams. Gave lectures on topics ranging from developmental physiology to biostatistics.

Refereeing

Algebra & Number Theory

Honors and Awards

2021 NSF Graduate Research Fellow
2021 UChicago Crerar Fellow
2014 International Biology Olympiad - Silver Medal (25th)
2015 North American Computational Linguistics Olympiad - U.S. Alternate (11th)